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LITTLEWOOD'S MUTUALLY TOUCHING CYLINDERS: THE MAXIMUM IS SEVEN

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Abstract. We give a computer-assisted proof that eight congruent infinite circular cylinders in $\mathbb{R}^3$ cannot touch pairwise. Since seven such cylinders exist, the maximum number of congruent mutually touching infinite cylinders is seven. This answers a question of Littlewood from 1968.

1. Introduction

1.1. The result.

Theorem 1.1. *There are no eight lines in $\mathbb{R}^3$ that are pairwise at the same positive distance.*

1.2. History of the problem.

Littlewood [8] asked for the maximum number of congruent infinite circular cylinders in $\mathbb{R}^3$ so that every pair is touching (see also [4, Section 2.4, Problem 14]). The problem is equivalent to determining the maximum number of lines in $\mathbb{R}^3$ at a common pairwise distance.

Bezdek [1] proved an upper bound of 24 in 2005. Bozóki, Lee and Rónyai [3] constructed a configuration of seven touching cylinders in 2015. Pikhitsa and Choi [11] attached to every pair of touching cylinders a sign, which they called chirality, and Pikhitsa and Pikhitsa [12] observed that the matrix of these signs is a Seidel matrix. These are the signs of pairs used below. Koizumi [7] gave the first rigorous results about them and proved in 2025 that the maximum is at most 18. Dillon, Koizumi and Luo [5] reduced this bound to 10 in 2025. Höfer [6] improved it to 9 in 2026 by a quadratic form argument. In higher dimensions, Solymosi and Zahl [14] proved that the number of mutually touching congruent cylinders in $\mathbb{R}^d$ is finite.

1.3. Structure of the proof.

Two cylinders of radius r touch if and only if the shortest distance between their central lines is $2r$. Littlewood's problem is equivalent to finding the maximum number of pairwise equidistant lines in $\mathbb{R}^3$. Our proof reduces this geometric question to one about signs: we attach a sign to each pair of lines and to each triple of lines, then show that certain sign configurations on four or five lines are impossible.

Once these obstructions are established, the geometry drops out entirely. What remains is a finite combinatorial problem: assign a sign to every pair and every triple of $\{0, 1, \ldots, 7\}$ so that no forbidden configuration occurs. We prove that no such assignment exists.

There are 2^{84} assignments in total, approximately $2 \cdot 10^{25}$. A computer testing one billion per second would require roughly a billion years to exhaust them. The assignments form the leaves of a binary search tree. Each forbidden configuration involves only a few of the 84 signs, so it becomes detectable long before reaching a leaf. Once a forbidden configuration is detected, the entire subtree below is pruned. In the end, the search visits approximately $2 \cdot 10^6$ nodes and completes in two seconds [10].

2. Lines and their signs

2.1. **Vectors.** For $\overrightarrow{x}, \overrightarrow{y}, \overrightarrow{z} \in \mathbb{R}^3$ we write $\overrightarrow{x} \cdot \overrightarrow{y}$ for the dot product and $\overrightarrow{x} \times \overrightarrow{y}$ for the cross product. We write $[\overrightarrow{x}\ \overrightarrow{y}\ \overrightarrow{z}]$ for the determinant of the matrix with columns $\overrightarrow{x}, \overrightarrow{y}, \overrightarrow{z}$. It equals $\overrightarrow{x} \cdot (\overrightarrow{y} \times \overrightarrow{z})$. We use three standard facts. The cross product $\overrightarrow{x} \times \overrightarrow{y}$ is zero exactly when $\overrightarrow{x}$ and $\overrightarrow{y}$ are parallel. Its length is $\|\overrightarrow{x}\|\|\overrightarrow{y}\| \sin\theta$, where θ is the angle between $\overrightarrow{x}$ and $\overrightarrow{y}$. The determinant $[\overrightarrow{x}\ \overrightarrow{y}\ \overrightarrow{z}]$ is zero exactly when $\overrightarrow{x}, \overrightarrow{y}, \overrightarrow{z}$ lie in a common plane through the origin. We also use the identity

$$(\overrightarrow{x} \times \overrightarrow{y}) \times \overrightarrow{z} = \overrightarrow{y}\,(\overrightarrow{x} \cdot \overrightarrow{z}) - \overrightarrow{x}\,(\overrightarrow{y} \cdot \overrightarrow{z}). \tag{1}$$

2.2. **Coordinates of a line.** An *oriented line* ℓ is given by a point $\overrightarrow{p}$ and a unit vector $\overrightarrow{u}$. It is the set $\{\overrightarrow{p} + t\overrightarrow{u} : t \in \mathbb{R}\}$. We call $\overrightarrow{u}$ the *direction* of ℓ. The *moment* of ℓ is the vector

$$\overrightarrow{w} = \overrightarrow{p} \times \overrightarrow{u}.$$

The moment does not depend on which point $\overrightarrow{p}$ of ℓ is used. Indeed $(\overrightarrow{p} + t\overrightarrow{u}) \times \overrightarrow{u} = \overrightarrow{p} \times \overrightarrow{u}$. So the pair $(\overrightarrow{u}, \overrightarrow{w})$ is determined by the oriented line. It satisfies $\|\overrightarrow{u}\| = 1$ and $\overrightarrow{u} \cdot \overrightarrow{w} = 0$. Conversely, let $(\overrightarrow{u}, \overrightarrow{w})$ be any pair with $\|\overrightarrow{u}\| = 1$ and $\overrightarrow{u} \cdot \overrightarrow{w} = 0$. Then $(\overrightarrow{u}, \overrightarrow{w})$ is the pair of the line through the point $\overrightarrow{u} \times \overrightarrow{w}$ with direction $\overrightarrow{u}$. Indeed $(\overrightarrow{u} \times \overrightarrow{w}) \times \overrightarrow{u} = \overrightarrow{w}\,(\overrightarrow{u} \cdot \overrightarrow{u}) - \overrightarrow{u}\,(\overrightarrow{w} \cdot \overrightarrow{u}) = \overrightarrow{w}$ by (1). Reversing the orientation of ℓ replaces $(\overrightarrow{u}, \overrightarrow{w})$ by $(-\overrightarrow{u}, -\overrightarrow{w})$.

We write $\ell = (\overrightarrow{u}, \overrightarrow{w})$.

2.3. **The Klein product.** For oriented lines $\ell = (\overrightarrow{u}, \overrightarrow{w})$ and $\ell' = (\overrightarrow{u}', \overrightarrow{w}')$ put

$$\mathrm{Kl}(\ell, \ell') = \overrightarrow{u} \cdot \overrightarrow{w}' + \overrightarrow{u}' \cdot \overrightarrow{w}.$$

This number is symmetric in ℓ and ℓ'. For a single line, $\mathrm{Kl}(\ell, \ell) = 2\,\overrightarrow{u} \cdot \overrightarrow{w} = 0$.

Lemma 2.1. *Let $\ell = (\overrightarrow{u}, \overrightarrow{p} \times \overrightarrow{u})$ and $\ell' = (\overrightarrow{u}', \overrightarrow{p}' \times \overrightarrow{u}')$ be oriented lines.*

(a) $\mathrm{Kl}(\ell, \ell') = (\overrightarrow{p} - \overrightarrow{p}') \cdot (\overrightarrow{u} \times \overrightarrow{u}')$.

(b) If $\overrightarrow{u}$ and $\overrightarrow{u}'$ are not parallel, then

$$\mathrm{dist}(\ell, \ell') = \frac{|(\overrightarrow{p} - \overrightarrow{p}') \cdot (\overrightarrow{u} \times \overrightarrow{u}')|}{\|\overrightarrow{u} \times \overrightarrow{u}'\|} = \frac{|\,\mathrm{Kl}(\ell, \ell')|}{\|\overrightarrow{u} \times \overrightarrow{u}'\|}.$$

(c) If $\overrightarrow{u}$ and $\overrightarrow{u}'$ are not parallel, then $\mathrm{dist}(\ell, \ell') = 1$ *if and only if* $\mathrm{Kl}(\ell, \ell') = \pm\|\overrightarrow{u} \times \overrightarrow{u}'\|$.

Proof. (a) We have $\overrightarrow{u} \cdot (\overrightarrow{p}' \times \overrightarrow{u}') = [\overrightarrow{u}\ \overrightarrow{p}'\ \overrightarrow{u}'] = [\overrightarrow{p}'\ \overrightarrow{u}'\ \overrightarrow{u}] = \overrightarrow{p}' \cdot (\overrightarrow{u}' \times \overrightarrow{u}) = -\overrightarrow{p}' \cdot (\overrightarrow{u} \times \overrightarrow{u}')$. Likewise $\overrightarrow{u}' \cdot (\overrightarrow{p} \times \overrightarrow{u}) = \overrightarrow{p} \cdot (\overrightarrow{u} \times \overrightarrow{u}')$. Adding the two gives (a).

The first equality in (b) is the standard formula for the distance between two skew lines. The second follows from (a). Part (c) follows from (b). □

Lemma 2.2. *Let $\ell_1, \ldots, \ell_n$ be oriented lines with $\ell_i = (\overrightarrow{u}_i, \overrightarrow{w}_i)$. Put $\ell_i' = (\overrightarrow{u}_i, -\overrightarrow{w}_i)$. Then $\ell_1', \ldots, \ell_n'$ are oriented lines,* $\mathrm{dist}(\ell_i', \ell_j') = \mathrm{dist}(\ell_i, \ell_j)$ *for all i, j, and* $\mathrm{Kl}(\ell_i', \ell_j') = -\mathrm{Kl}(\ell_i, \ell_j)$ *for all i, j.*

Proof. The pair $(\overrightarrow{u}_i, -\overrightarrow{w}_i)$ satisfies $\|\overrightarrow{u}_i\| = 1$ and $\overrightarrow{u}_i \cdot (-\overrightarrow{w}_i) = 0$. So it is an oriented line. Let ℓ_i pass through $\overrightarrow{p}_i$. The point reflection $\overrightarrow{x} \mapsto -\overrightarrow{x}$ maps ℓ_i onto the line through $-\overrightarrow{p}_i$ with direction $\overrightarrow{u}_i$. The moment of that line is $(-\overrightarrow{p}_i) \times \overrightarrow{u}_i = -\overrightarrow{w}_i$. So ℓ_i' is the image of ℓ_i under the point reflection. Point reflections preserve distances. Finally, $\mathrm{Kl}(\ell_i', \ell_j') = \overrightarrow{u}_i \cdot (-\overrightarrow{w}_j) + \overrightarrow{u}_j \cdot (-\overrightarrow{w}_i) = -\mathrm{Kl}(\ell_i, \ell_j)$. □

2.4. **Touching families.** A *touching family* is a set of at least five lines in $\mathbb{R}^3$ that are pairwise at distance 1. We quote one theorem from the literature.

Theorem 2.3 (Höfer)**.** *Let $\ell_1, \dots, \ell_n$ be lines in $\mathbb{R}^3$ with $n \geq 5$ that are pairwise at the same positive distance. Then no two of the lines are parallel.*

Proof. See [6, Lemma 1(a)] or [7]. □

Lemma 2.4 ([6, Lemma 1(b)])**.** *Let ℓ_1, ℓ_2, ℓ_3 be pairwise non-parallel lines at pairwise distance 1, with directions $\vec{u}_1, \vec{u}_2, \vec{u}_3$. Then $[\vec{u}_1\, \vec{u}_2\, \vec{u}_3] \neq 0$.*

Proof. Suppose $[\vec{u}_1\, \vec{u}_2\, \vec{u}_3] = 0$. Then there is a unit vector $\vec{n}$ with $\vec{u}_1, \vec{u}_2, \vec{u}_3 \perp \vec{n}$. Each cross product $\vec{u}_i \times \vec{u}_j$ is parallel to $\vec{n}$. So $\vec{u}_i \times \vec{u}_j = \pm\|\vec{u}_i \times \vec{u}_j\|\, \vec{n}$. Let $\vec{p}_i$ be a point of ℓ_i and put $h_i = \vec{p}_i \cdot \vec{n}$. By Lemma 2.1(b),

$$\operatorname{dist}(\ell_i, \ell_j) = \frac{|(\vec{p}_i - \vec{p}_j) \cdot (\vec{u}_i \times \vec{u}_j)|}{\|\vec{u}_i \times \vec{u}_j\|} = |h_i - h_j|.$$

So the three numbers h_1, h_2, h_3 have all pairwise differences equal to ± 1. This is impossible. □

Let $\ell_1, \dots, \ell_n$ be a touching family. By Theorem 2.3 and Lemma 2.4 we have the following for all distinct indices $i, j, k \in \{1, \dots, n\}$:

$$\vec{u}_i \times \vec{u}_j \neq \vec{0}, \qquad \operatorname{Kl}(\ell_i, \ell_j) \neq 0, \qquad [\vec{u}_i\, \vec{u}_j\, \vec{u}_k] \neq 0. \tag{2}$$

These facts are used throughout without further mention.

2.5. **The sign data.** Let $\ell_1, \dots, \ell_n$ be a touching family with chosen orientations, where $\ell_i = (\vec{u}_i, \vec{w}_i)$. We attach two kinds of signs to it. For distinct indices $i, j \in \{1, \dots, n\}$ put

$$\sigma_{ij} = \operatorname{sgn} \operatorname{Kl}(\ell_i, \ell_j) \in \{-1, +1\}.$$

For distinct indices $a, b, c \in \{1, \dots, n\}$ put

$$\chi(a, b, c) = \operatorname{sgn}[\vec{u}_a\, \vec{u}_b\, \vec{u}_c] \in \{-1, +1\}.$$

Both signs are nonzero by (2). The sign σ_{ij} is symmetric in i, j. The sign $\chi(a, b, c)$ changes sign when two of its arguments are exchanged.

By Lemma 2.1(c),

$$\operatorname{Kl}(\ell_i, \ell_j) = \sigma_{ij}\|\vec{u}_i \times \vec{u}_j\| \qquad (i \neq j). \tag{3}$$

Let $m \in \{1, \dots, n\}$. Suppose that the orientation of the single line ℓ_m is reversed and the other lines are kept. Then $\vec{u}_m$ and $\vec{w}_m$ are replaced by $-\vec{u}_m$ and $-\vec{w}_m$. Since $\operatorname{Kl}(\ell_m, \ell_j) = \vec{u}_m \cdot \vec{w}_j + \vec{u}_j \cdot \vec{w}_m$, the sign σ_{mj} changes for every $j \neq m$. By multilinearity of the determinant, the sign $\chi(a, b, c)$ changes for every triple a, b, c that contains the index m. All other signs are unchanged.

Remark 2.5. The signs σ_{ij} are the chiralities of [11, 12, 7]. The Klein product, and formula (3) for it, appear in [7] and [5] as a symmetric bilinear form of signature $(3, 3)$ on $\mathbb{R}^6$. The signs $\chi(a, b, c)$ form the chirotope of the vector configuration $\vec{u}_1, \dots, \vec{u}_n$ in the sense of oriented matroid theory [2]. We do not use any theory of chirotopes. We use only the two identities of the next subsection.

2.6. **Two determinant identities.**

Lemma 2.6. *Let $\overrightarrow{u}_a, \overrightarrow{u}_b, \overrightarrow{u}_c, \overrightarrow{u}_d \in \mathbb{R}^3$ be vectors such that no three of them lie in a common plane through the origin. Put*

$$k_a = [\overrightarrow{u}_b\ \overrightarrow{u}_c\ \overrightarrow{u}_d], \quad k_b = -[\overrightarrow{u}_a\ \overrightarrow{u}_c\ \overrightarrow{u}_d], \quad k_c = [\overrightarrow{u}_a\ \overrightarrow{u}_b\ \overrightarrow{u}_d], \quad k_d = -[\overrightarrow{u}_a\ \overrightarrow{u}_b\ \overrightarrow{u}_c].$$

Then all four numbers are nonzero, and $k_a\overrightarrow{u}_a + k_b\overrightarrow{u}_b + k_c\overrightarrow{u}_c + k_d\overrightarrow{u}_d = \overrightarrow{0}$. Every vector $(c_a, c_b, c_c, c_d) \in \mathbb{R}^4$ with $c_a\overrightarrow{u}_a + c_b\overrightarrow{u}_b + c_c\overrightarrow{u}_c + c_d\overrightarrow{u}_d = \overrightarrow{0}$ is a scalar multiple of (k_a, k_b, k_c, k_d).

Proof. The four numbers are nonzero by assumption. Let $\overrightarrow{x} \in \mathbb{R}^3$ be arbitrary. Consider the 4×4 matrix M whose rows are $(\overrightarrow{x}\cdot\overrightarrow{u}_i,\ \overrightarrow{u}_i^{\mathsf{T}})$ for $i = a, b, c, d$. The first column of M is a linear combination of the other three columns, with coefficients the coordinates of $\overrightarrow{x}$. So $\det M = 0$. Expanding $\det M$ along the first column gives

$$\det M = (\overrightarrow{x}\cdot\overrightarrow{u}_a)k_a + (\overrightarrow{x}\cdot\overrightarrow{u}_b)k_b + (\overrightarrow{x}\cdot\overrightarrow{u}_c)k_c + (\overrightarrow{x}\cdot\overrightarrow{u}_d)k_d = \overrightarrow{x}\cdot(k_a\overrightarrow{u}_a + k_b\overrightarrow{u}_b + k_c\overrightarrow{u}_c + k_d\overrightarrow{u}_d).$$

Since this vanishes for every $\overrightarrow{x}$, the vector $k_a\overrightarrow{u}_a + k_b\overrightarrow{u}_b + k_c\overrightarrow{u}_c + k_d\overrightarrow{u}_d$ is zero. The vectors $\overrightarrow{u}_b, \overrightarrow{u}_c, \overrightarrow{u}_d$ form a basis of $\mathbb{R}^3$. So $\overrightarrow{u}_a$ has a unique expression as a combination of them. Hence the space of dependencies among $\overrightarrow{u}_a, \overrightarrow{u}_b, \overrightarrow{u}_c, \overrightarrow{u}_d$ has dimension one. □

Lemma 2.7. *Let $\overrightarrow{a}, \overrightarrow{v}_1, \overrightarrow{v}_2, \overrightarrow{v}_3, \overrightarrow{v}_4 \in \mathbb{R}^3$ and put $\Delta_{ij} = [\overrightarrow{a}\ \overrightarrow{v}_i\ \overrightarrow{v}_j]$. Then*

$$\Delta_{12}\Delta_{34} - \Delta_{13}\Delta_{24} + \Delta_{14}\Delta_{23} = 0. \tag{4}$$

Proof. If $\overrightarrow{a} = \overrightarrow{0}$, then all Δ_{ij} are zero. Otherwise choose vectors $\overrightarrow{e}_1, \overrightarrow{e}_2$ such that $\overrightarrow{e}_1, \overrightarrow{e}_2, \overrightarrow{a}$ form a basis of $\mathbb{R}^3$. Write $\overrightarrow{v}_i = f_i\overrightarrow{e}_1 + g_i\overrightarrow{e}_2 + h_i\overrightarrow{a}$ with real numbers f_i, g_i, h_i. The terms $h_i\overrightarrow{a}$ do not change any determinant Δ_{ij}. So

$$\Delta_{ij} = [\overrightarrow{a},\ f_i\overrightarrow{e}_1 + g_i\overrightarrow{e}_2,\ f_j\overrightarrow{e}_1 + g_j\overrightarrow{e}_2] = (f_ig_j - f_jg_i)\,[\overrightarrow{a}\ \overrightarrow{e}_1\ \overrightarrow{e}_2].$$

We can now factor $[\overrightarrow{a}\ \overrightarrow{e}_1\ \overrightarrow{e}_2]^2$ from the left-hand side of (4) and directly verify that the remaining terms cancel out after the expansion. □

3. Four sign lemmas

Throughout this section, $\ell_1, \ldots, \ell_n$ is a touching family with chosen orientations, with $\ell_i = (\overrightarrow{u}_i, \overrightarrow{w}_i)$ and with signs σ_{ij} and $\chi(a, b, c)$ as in Section 2.5.

3.1. **A bilinear identity.** The next lemma is a reformulation of an observation in [7] and [5]: the matrix $\big(\mathrm{Kl}(\ell_i, \ell_j)\big)$ is the Gram matrix of the vectors $(\overrightarrow{u}_i, \overrightarrow{w}_i) \in \mathbb{R}^6$ with respect to a bilinear form of signature $(3, 3)$.

Lemma 3.1. *Let $c = (c_1, \ldots, c_n)$ and $d = (d_1, \ldots, d_n)$ be in $\mathbb{R}^n$ and satisfy $\sum_i c_i\overrightarrow{u}_i = \overrightarrow{0}$ and $\sum_j d_j\overrightarrow{u}_j = \overrightarrow{0}$. Then*

$$\sum_{i\neq j} c_id_j\,\mathrm{Kl}(\ell_i, \ell_j) = 0. \tag{5}$$

Proof. The terms with $i = j$ may be added to the sum, because $\mathrm{Kl}(\ell_i, \ell_i) = 2\,\overrightarrow{u}_i\cdot\overrightarrow{w}_i = 0$. By the definition $\mathrm{Kl}(\ell_i, \ell_j) = \overrightarrow{u}_i\cdot\overrightarrow{w}_j + \overrightarrow{u}_j\cdot\overrightarrow{w}_i$,

$$\sum_{i,j} c_id_j\,\mathrm{Kl}(\ell_i, \ell_j) = \Big(\sum_i c_i\overrightarrow{u}_i\Big)\cdot\Big(\sum_j d_j\overrightarrow{w}_j\Big) + \Big(\sum_j d_j\overrightarrow{u}_j\Big)\cdot\Big(\sum_i c_i\overrightarrow{w}_i\Big) = 0 + 0 = 0. \quad \square$$

3.2. **Four lines.** Let Q be a set of four indices from $\{1,\dots,n\}$. Write $Q=\{a,b,c,d\}$ with $a<b<c<d$ and define $k^Q=(k^Q_a,k^Q_b,k^Q_c,k^Q_d)$ by Lemma 2.6:

$$k^Q_a=[\vec{u}_b\,\vec{u}_c\,\vec{u}_d],\quad k^Q_b=-[\vec{u}_a\,\vec{u}_c\,\vec{u}_d],\quad k^Q_c=[\vec{u}_a\,\vec{u}_b\,\vec{u}_d],\quad k^Q_d=-[\vec{u}_a\,\vec{u}_b\,\vec{u}_c]. \tag{6}$$

Then $\sum_{i\in Q}k^Q_i\vec{u}_i=\vec{0}$, all entries are nonzero, and the sign of each entry is read off from χ.

Lemma 3.2. *Let Q be a set of four indices and put*

$$\varepsilon_{ij}=\sigma_{ij}\,\operatorname{sgn}(k^Q_ik^Q_j)\qquad\text{for the six pairs }(i,j)\text{ with }i,j\in Q,\ i<j.$$

Let P be the set of pairs with $\varepsilon_{ij}=+1$. Then one of the following holds.

(i) $|P|=2$ and the two pairs in P have a common element.
(ii) $|P|=3$.
(iii) $|P|=4$ and the two pairs not in P have a common element.

Proof. Put $\vec{v}_i=k^Q_i\vec{u}_i$ for $i\in Q$. Then $\sum_{i\in Q}\vec{v}_i=\vec{0}$. Put $A_{ij}=\|\vec{v}_i\times\vec{v}_j\|$. It is positive, and $A_{ij}=A_{ji}$.

Apply Lemma 3.1 with $c=d=k^Q$, extended by zeros to the indices outside Q. By (3),

$$k^Q_ik^Q_j\,\mathrm{Kl}(\ell_i,\ell_j)=\sigma_{ij}\,k^Q_ik^Q_j\,\|\vec{u}_i\times\vec{u}_j\|=\varepsilon_{ij}\,|k^Q_i|\,|k^Q_j|\,\|\vec{u}_i\times\vec{u}_j\|=\varepsilon_{ij}A_{ij}.$$

Hence

$$\sum_{i<j}\varepsilon_{ij}A_{ij}=0,\qquad i,j\in Q. \tag{7}$$

Fix $i\in Q$. The three vectors $\vec{v}_i\times\vec{v}_j$ with $j\in Q\setminus\{i\}$ sum to $\vec{v}_i\times(-\vec{v}_i)=\vec{0}$. Each is nonzero. No two of them are parallel. Indeed, if $\vec{v}_i\times\vec{v}_j$ and $\vec{v}_i\times\vec{v}_k$ were parallel, then both would be orthogonal to $\vec{u}_i$, $\vec{u}_j$ and $\vec{u}_k$. Then $\vec{u}_i$, $\vec{u}_j$, $\vec{u}_k$ would lie in a common plane, contradicting (2). So $\vec{v}_i\times\vec{v}_j=-(\vec{v}_i\times\vec{v}_k)-(\vec{v}_i\times\vec{v}_l)$ is a sum of two non-parallel vectors, and the triangle inequality is strict:

$$A_{ij}<A_{ik}+A_{il}\qquad\text{whenever }\{i,j,k,l\}=Q. \tag{8}$$

We now rule out the other possibilities for P.

If $|P|=0$, the left side of (7) is negative.

If $|P|=1$, say $P=\{(a,b)\}$, then (7) gives $A_{ab}=A_{ac}+A_{ad}+A_{bc}+A_{bd}+A_{cd}$. By (8), $A_{ab}<A_{ac}+A_{ad}$. These contradict each other.

If $|P|=2$ and the two pairs are disjoint, say $P=\{(a,b),(c,d)\}$, then (7) gives $A_{ab}+A_{cd}=A_{ac}+A_{ad}+A_{bc}+A_{bd}$. By (8) for $i=a$ and for $i=b$, we get $A_{ab}<A_{ac}+A_{ad}$ and $A_{ab}<A_{bc}+A_{bd}$. Adding, $2A_{ab}<A_{ac}+A_{ad}+A_{bc}+A_{bd}$. In the same way, (8) for $i=c$ and $i=d$ gives $2A_{cd}<A_{ac}+A_{bc}+A_{ad}+A_{bd}$. Adding the last two inequalities gives $A_{ab}+A_{cd}<A_{ac}+A_{ad}+A_{bc}+A_{bd}$, which contradicts the equation.

The cases $|P|=6$, $|P|=5$, and $|P|=4$ with disjoint negative pairs follow by the same arguments with all signs ε_{ij} reversed. Equation (7) is symmetric under this reversal. □

Remark 3.3. The proof above is the template for the two lemmas that follow. The positions of the lines enter only through Lemma 3.1. Equation (3) then turns each $\mathrm{Kl}(\ell_i,\ell_j)$ into a sign times the area $\|\vec{u}_i\times\vec{u}_j\|$. From then on only the directions $\vec{u}_i$ and the signs σ_{ij} matter, and the argument is about triangle inequalities between areas.

Up to relabelling there are eleven ways to assign signs to the six pairs of four lines. The lemma allows five of them. In terms of the graph of negative pairs, the allowed graphs are: two edges with a common vertex, any three edges, and four edges whose complement is two edges with a common vertex.

3.3. **Two four-element subsets.**

Lemma 3.4. *Let I and J be sets of four indices from $\{1,\dots,n\}$. For each $g\in\{-1,+1\}$ there exists $j\in J$ such that at least two indices $i\in I\setminus\{j\}$ satisfy*

$$\sigma_{ji}\,\operatorname{sgn}(k^J_j k^I_i) = g.$$

Proof. Put $c=k^I$ and $d=k^J$, extended by zeros to the indices outside I and J. By (5),

$$\sum_{j\in J}\sum_{i\in I,\,i\neq j} g\,d_jc_i\,\mathrm{Kl}(\ell_j,\ell_i)=0. \tag{9}$$

Fix $j\in J$. For $i\in I$ put $\overrightarrow{y}_i = c_i\,\overrightarrow{u}_j\times\overrightarrow{u}_i$. Then $\sum_{i\in I}\overrightarrow{y}_i = \overrightarrow{u}_j\times\sum_{i\in I}c_i\overrightarrow{u}_i=\overrightarrow{0}$. The vector $\overrightarrow{y}_i$ is zero exactly when $i=j$. Two nonzero vectors $\overrightarrow{y}_i$, $\overrightarrow{y}_{i'}$ are not parallel, since otherwise $\overrightarrow{u}_j,\overrightarrow{u}_i,\overrightarrow{u}_{i'}$ would lie in a common plane. By (3), for $i\neq j$,

$$g\,d_jc_i\,\mathrm{Kl}(\ell_j,\ell_i) = g\,\sigma_{ji}\operatorname{sgn}(d_jc_i)\,|d_j|\,\|\overrightarrow{y}_i\|.$$

This number is positive when $\sigma_{ji}\operatorname{sgn}(d_jc_i)=g$ and negative otherwise.

Suppose that at most one index $i\in I\setminus\{j\}$ has $\sigma_{ji}\operatorname{sgn}(d_jc_i)=g$. If there is no such index, then the inner sum in (9) has at least three negative terms and no positive ones. If there is exactly one such index, call it k, then the inner sum equals

$$|d_j|\Big(\|\overrightarrow{y}_k\| - \sum_{i\in I,\,i\neq k}\|\overrightarrow{y}_i\|\Big),$$

which is negative because $\overrightarrow{y}_k=-\sum_{i\neq k}\overrightarrow{y}_i$ is a sum of at least two nonzero non-parallel vectors, so the triangle inequality is strict. In both cases the inner sum in (9) is negative.

If the conclusion of the lemma failed, the inner sum would be negative for every $j\in J$. Summing over j would contradict (9). □

3.4. **Five lines.**

Lemma 3.5. *Let $s\in\{-1,+1\}$. There are no five lines $\ell_0,\ell_1,\ell_2,\ell_3,\ell_4$ of the family such that*

(i) $\chi(0,1,2)=\chi(0,1,3)=\chi(0,1,4)=\chi(1,2,3)=\chi(1,2,4)=+1$ *and* $\chi(0,2,3)=\chi(0,3,4)=\chi(1,3,4)=-1$, *and*

(ii) $\sigma_{03}=-s$ *and* $\sigma_{ij}=s$ *for the eight pairs*

$$(i,j)\in\{(0,1),(0,2),(0,4),(1,2),(1,3),(1,4),(2,3),(2,4)\}.$$

The signs σ_{34}, $\chi(0,2,4)$ and $\chi(2,3,4)$ are not restricted by the hypotheses.

Proof. Suppose such lines exist. Write $[abc]$ for $[\overrightarrow{u}_a\,\overrightarrow{u}_b\,\overrightarrow{u}_c]$. Put

$$A=[123],\quad B=-[023],\quad C=[013],\quad D=[012],\quad P=[124],\quad Q=-[024],\quad R=[014].$$

By (i), the numbers A,B,C,D,P,R are positive. Lemma 2.7 with $\overrightarrow{a}=\overrightarrow{u}_0$ gives $[012][034]-[013][024]+[014][023]=0$. Hence

$$C\,[024]=[013][024]=[012][034]+[014][023]=D[034]-RB.$$

Both terms on the right are negative by (i). So $[024]<0$, and $Q>0$.

By Lemma 2.6 applied to $\overrightarrow{u}_0,\overrightarrow{u}_1,\overrightarrow{u}_2,\overrightarrow{u}_3$ and to $\overrightarrow{u}_0,\overrightarrow{u}_1,\overrightarrow{u}_2,\overrightarrow{u}_4$,

$$A\overrightarrow{u}_0+B\overrightarrow{u}_1+C\overrightarrow{u}_2=D\overrightarrow{u}_3,\qquad P\overrightarrow{u}_0+Q\overrightarrow{u}_1+R\overrightarrow{u}_2=D\overrightarrow{u}_4. \tag{10}$$

We compute $[134]$ and $[034]$ from (10). Substituting the expressions for $\overrightarrow{u}_3$ and $\overrightarrow{u}_4$ and expanding,

$$D^2[134]=[\overrightarrow{u}_1,\,A\overrightarrow{u}_0+C\overrightarrow{u}_2,\,P\overrightarrow{u}_0+R\overrightarrow{u}_2]=AR[102]+CP[120]=(CP-AR)D,$$

$$D^2[034] = [\vec{u}_0,\, B\vec{u}_1 + C\vec{u}_2,\, Q\vec{u}_1 + R\vec{u}_2] = BR[012] + CQ[021] = (BR - CQ)D.$$

So $CP - AR = D[134] < 0$ and $CQ - BR = -D[034] > 0$. Put

$$x = \frac{CP}{AR}, \qquad y = \frac{CQ}{BR}.$$

Then $0 < x < 1$ and $y > 1$.

Now rescale the directions. Put $\vec{v}_0 = A\vec{u}_0$, $\vec{v}_1 = B\vec{u}_1$, $\vec{v}_2 = C\vec{u}_2$, $\vec{v}_3 = D\vec{u}_3$ and $\vec{v}_4 = (DC/R)\,\vec{u}_4$. All scale factors are positive. Multiplying the second equation in (10) by C/R and using the definitions of x and y, we obtain

$$\vec{v}_3 = \vec{v}_0 + \vec{v}_1 + \vec{v}_2, \qquad \vec{v}_4 = x\,\vec{v}_0 + y\,\vec{v}_1 + \vec{v}_2. \tag{11}$$

Write $\vec{v}_i = s_i\vec{u}_i$ with $s_i > 0$ and put $K_{ij} = s\,s_i s_j\,\mathrm{Kl}(\ell_i, \ell_j)$. By (3) and (ii),

$$K_{ij} = s\,\sigma_{ij}\|\vec{v}_i \times \vec{v}_j\| = \begin{cases} \|\vec{v}_i \times \vec{v}_j\| & \text{for the eight pairs in (ii),} \\ -\|\vec{v}_0 \times \vec{v}_3\| & \text{for } (i,j) = (0,3). \end{cases}$$

By (11),

$$\vec{v}_0 + \vec{v}_1 + \vec{v}_2 - \vec{v}_3 = \vec{0} \qquad \text{and} \qquad x\vec{v}_0 + y\vec{v}_1 + \vec{v}_2 - \vec{v}_4 = \vec{0}.$$

Since $\vec{v}_i$ is a positive multiple of $\vec{u}_i$, these are two linear combinations of $\vec{u}_0, \dots, \vec{u}_4$ that are zero. Apply (5) to each of them, with $c = d$, and multiply both sides by s. Collecting the coefficients of K_{ij} gives

$$K_{01} + K_{02} + K_{12} - K_{03} - K_{13} - K_{23} = 0, \tag{12}$$

$$xyK_{01} + xK_{02} + yK_{12} - xK_{04} - yK_{14} - K_{24} = 0. \tag{13}$$

Put $\vec{a} = \vec{v}_1 \times \vec{v}_2$, $\vec{b} = \vec{v}_2 \times \vec{v}_0$ and $\vec{c} = \vec{v}_0 \times \vec{v}_1$. From (11),

$$\vec{v}_0 \times \vec{v}_3 = \vec{c} - \vec{b}, \qquad \vec{v}_1 \times \vec{v}_3 = \vec{a} - \vec{c}, \qquad \vec{v}_2 \times \vec{v}_3 = \vec{b} - \vec{a},$$
$$\vec{v}_0 \times \vec{v}_4 = y\vec{c} - \vec{b}, \qquad \vec{v}_1 \times \vec{v}_4 = \vec{a} - x\vec{c}, \qquad \vec{v}_2 \times \vec{v}_4 = x\vec{b} - y\vec{a}.$$

Substituting these into (12) and (13) gives

$$\|\vec{a}\| + \|\vec{b}\| + \|\vec{c}\| + \|\vec{b} - \vec{c}\| - \|\vec{a} - \vec{c}\| - \|\vec{a} - \vec{b}\| = 0, \tag{14}$$

$$y\|\vec{a}\| + x\|\vec{b}\| + xy\|\vec{c}\| - x\|y\vec{c} - \vec{b}\| - y\|\vec{a} - x\vec{c}\| - \|x\vec{b} - y\vec{a}\| = 0. \tag{15}$$

Let L_1 and L_2 be the left-hand sides of (14) and (15). Put

$$T_1 = \|\vec{a} - x\vec{c}\| + (1-x)\|\vec{c}\| - \|\vec{a} - \vec{c}\|,$$
$$T_2 = \|y\vec{a} - x\vec{b}\| + (y-x)\|\vec{b}\| - y\|\vec{a} - \vec{b}\|.$$

Both are nonnegative by the triangle inequality, since $\vec{a} - \vec{c} = (\vec{a} - x\vec{c}) - (1-x)\vec{c}$ and $y(\vec{a} - \vec{b}) = (y\vec{a} - x\vec{b}) - (y-x)\vec{b}$, and since $0 < x < 1 < y$. A direct expansion shows

$$yL_1 - L_2 = y\|\vec{b} - \vec{c}\| + x\|y\vec{c} - \vec{b}\| + y\,T_1 + T_2.$$

The left side is zero by (14) and (15). Every term on the right is nonnegative. The term $y\|\vec{b} - \vec{c}\|$ is positive, because $\vec{b} - \vec{c} = -(\vec{v}_0 \times \vec{v}_3) \neq \vec{0}$. This is a contradiction. □

4. Reduction to a finite statement

4.1. **Sign patterns.** Let V be a finite set. A *sign pattern* on V is a pair (σ, χ), where σ assigns a sign $\sigma_{ij} = \sigma_{ji} \in \{-1, +1\}$ to every pair of distinct elements $i, j \in V$, and χ assigns a sign $\chi(a, b, c) \in \{-1, +1\}$ to every triple of distinct elements $a, b, c \in V$, with χ changing sign under the exchange of two arguments.

For a four-element subset $Q = \{a < b < c < d\}$ of V we define signs $\kappa_i^Q \in \{-1, +1\}$ for $i \in Q$ by

$$\kappa_a^Q = \chi(b, c, d), \quad \kappa_b^Q = -\chi(a, c, d), \quad \kappa_c^Q = \chi(a, b, d), \quad \kappa_d^Q = -\chi(a, b, c).$$

These are the signs of the entries of k^Q in (6) when χ comes from a touching family.

Definition 4.1. A sign pattern (σ, χ) on V is *admissible* if it has the following four properties.

(P1) For every $a \in V$ and all distinct $v_1, v_2, v_3, v_4 \in V \setminus \{a\}$, the three signs

$$\chi(a, v_1, v_2)\chi(a, v_3, v_4), \qquad -\chi(a, v_1, v_3)\chi(a, v_2, v_4), \qquad \chi(a, v_1, v_4)\chi(a, v_2, v_3)$$

are not all equal.

(P2) For every four-element subset $Q \subseteq V$, the set P of pairs (i, j) with $i, j \in Q$, $i < j$ and $\sigma_{ij}\kappa_i^Q\kappa_j^Q = +1$ satisfies (i), (ii) or (iii) of Lemma 3.2.

(P3) For every five-element subset $S \subseteq V$, all four-element subsets $I, J \subseteq S$, and every $g \in \{-1, +1\}$, there is $j \in J$ such that at least two indices $i \in I \setminus \{j\}$ satisfy $\sigma_{ji}\,\kappa_j^J\kappa_i^I = g$.

(P4) For every injective map $p : \{0, 1, 2, 3, 4\} \to V$, every $f = (f_0, \ldots, f_4) \in \{-1, +1\}^5$ and every $s \in \{-1, +1\}$, the pattern (σ', χ') on $\{0, \ldots, 4\}$ defined by

$$\sigma'_{ab} = f_a f_b\, \sigma_{p(a)p(b)}, \qquad \chi'(a, b, c) = f_a f_b f_c\, \chi(p(a), p(b), p(c))$$

does not satisfy both (i) and (ii) of Lemma 3.5.

4.2. **The finite statement.**

Theorem 4.2. *There is no admissible sign pattern on an eight-element set.*

Theorem 4.2 is proved by computer in Section 5.

Proof of Theorem 1.1. Suppose $\ell_1, \ldots, \ell_8$ are lines in $\mathbb{R}^3$ that are pairwise at the same distance $\delta > 0$. Scale $\mathbb{R}^3$ by the factor $1/\delta$. The scaled lines are a touching family. Choose orientations. Let (σ, χ) be the sign pattern on $V = \{1, \ldots, 8\}$ defined in Section 2.5. We show that it is admissible. This contradicts Theorem 4.2.

For (P1), apply Lemma 2.7 to $\overrightarrow{a} = \overrightarrow{u}_a$ and $\overrightarrow{v}_i = \overrightarrow{u}_{v_i}$. The three products in (P1) are the signs of the three terms of (4). These terms are nonzero by (2), and three nonzero numbers with sum zero cannot all have the same sign.

(P2) holds by Lemma 3.2, because $\kappa_i^Q = \operatorname{sgn} k_i^Q$.

(P3) holds by Lemma 3.4, because $\kappa_j^J\kappa_i^I = \operatorname{sgn}(k_j^J k_i^I)$.

For (P4), let p, f and s be given. Consider the five lines $\ell_{p(0)}, \ldots, \ell_{p(4)}$, where the orientation of $\ell_{p(a)}$ is reversed if $f_a = -1$. These are five lines of a touching family. Reversing the orientation of a line multiplies its direction and its moment by -1. So the signs of this reoriented subfamily are exactly σ' and χ'. By Lemma 3.5, they do not satisfy both (i) and (ii). □

4.3. **Symmetries.** The verification uses the symmetries of Definition 4.1 to reduce the number of cases.

Lemma 4.3. *Let (σ, χ) be an admissible sign pattern on V. Each of the following is again an admissible sign pattern on V.*

(a) Relabelling. *$(\sigma \circ \pi, \chi \circ \pi)$ for a bijection $\pi : V \to V$, where $(\sigma \circ \pi)_{ij} = \sigma_{\pi(i)\pi(j)}$ and $(\chi \circ \pi)(a,b,c) = \chi(\pi(a), \pi(b), \pi(c))$.*

(b) Reorientation. *Fix $m \in V$. Change the sign of σ_{mj} for all $j \neq m$, and change the sign of $\chi(a,b,c)$ for all triples a, b, c that contain m. Keep all other signs.*

(c) Reflection. *$(-\sigma, \chi)$.*

Proof. Property (P4) is invariant under all three operations by construction. The quantifiers over p, f and s absorb relabelling, reorientation and reflection, respectively. Property (P1) does not involve σ, is invariant under relabelling, and is invariant under reorientation because reorientation multiplies each of the three products by the same factor $f_a^2 f_{v_1} f_{v_2} f_{v_3} f_{v_4}$, where $f_i = -1$ if $i = m$ and $f_i = +1$ otherwise.

Consider the quantity $\sigma_{ij}\kappa_i^Q \kappa_j^Q$ in (P2). Under a relabelling, κ^Q is replaced by $\pm\kappa^Q$, because the ordering $a < b < c < d$ may change, and a product $\kappa_i^Q \kappa_j^Q$ is unchanged. Under the reorientation of m, the sign σ_{ij} changes exactly when $m \in \{i, j\}$, and κ_i^Q changes exactly when $m \in Q \setminus \{i\}$. If $m \notin Q$ nothing changes. If $m = i$, then σ_{ij} and κ_j^Q change and κ_i^Q does not. If $m \in Q \setminus \{i, j\}$, then κ_i^Q and κ_j^Q change and σ_{ij} does not. So the product is invariant. Under reflection the product changes sign. The conditions (i), (ii), (iii) of Lemma 3.2 are symmetric under exchanging the set P with its complement.

Consider the quantity $\sigma_{ji}\kappa_j^J \kappa_i^I$ in (P3). Relabelling changes κ^I and κ^J by global signs, which are absorbed by g. Under the reorientation of m, the factor σ_{ji} changes exactly when $m \in \{i, j\}$, the factor κ_j^J exactly when $m \in J \setminus \{j\}$, and the factor κ_i^I exactly when $m \in I \setminus \{i\}$. In each of the cases $m = j$, $m = i$ and $m \notin \{i, j\}$, the number of changing factors is congruent to $[m \in I] + [m \in J]$ modulo 2, where $[\,\cdot\,]$ is 1 if the statement holds and 0 otherwise. So all quantities $\sigma_{ji}\kappa_j^J \kappa_i^I$ with $j \in J$, $i \in I$ change by the same factor. This factor is absorbed by g. Reflection changes every quantity by the factor -1, which is also absorbed by g. □

Remark 4.4. The orbits of σ under relabelling and reorientation are called *two-graphs* or *switching classes* [13]. A triple $\{i, j, k\}$ is called odd if $\sigma_{ij}\sigma_{jk}\sigma_{ki} = -1$. The set of odd triples does not change under reorientation. It determines σ up to reorientation. There are 243 two-graphs on eight points up to relabelling [9]. Reflection maps the set of odd triples to its complement. The connection between chiralities and Seidel matrices was noted in [12], and switching is used in [5].

5. Verification of Theorem 4.2

The program is available at [10]. It is about five hundred lines of C++ and depends on nothing but the standard library. This section describes what it does.

5.1. **The cases.** Let $V = \{0, 1, \ldots, 7\}$. By Lemma 4.3(b), every sign pattern can be reoriented so that $\sigma_{0j} = +1$ for all $j \neq 0$, and admissibility is not affected. So it suffices to consider the 2^{21} assignments σ with $\sigma_{0j} = +1$ and arbitrary signs on the 21 pairs within $\{1, \ldots, 7\}$. The program encodes each of them as an integer between 0 and $2^{21} - 1$.

By Lemma 4.3(a), relabelling the points does not affect admissibility either. If σ' is obtained from σ by a permutation of V followed by the reorientation that restores $\sigma'_{0j} = +1$, then σ

admits an admissible χ if and only if σ' does. The program keeps a table with one bit for each of the 2^{21} assignments. It runs through them in increasing order. When it reaches an assignment σ whose bit is not yet set, it searches for a χ such that (σ, χ) is admissible, as described next, and then sets the bits of the images of σ under all 8! permutations of V. The program stops if a χ is found. At the end it checks that every bit is set.

5.2. **The search.** The unknowns are the 56 signs $\chi(a, b, c)$ with $a < b < c$. The value of χ on a triple in another order is the sign of the sorting permutation times the value on the sorted triple. The triples are numbered so that all triples inside $\{0, \dots, m\}$ come before any triple containing $m + 1$.

The triples are assigned one at a time, in this order, by depth-first search. Each condition in Definition 4.1 concerns a small set of triples: six triples for an instance of (P1), four for a four-element subset in (P2), and ten for a five-element subset in (P3) and (P4). The condition is checked as soon as the last of its triples is assigned. If it fails, the current value of that triple is rejected and the search backtracks. The list of conditions to check after each triple is computed once, before the search starts.

(P1), (P2) and (P3) are checked by evaluating their statements. For (P3) this means all 25 pairs (I, J) of four-element subsets of each five-element subset.

(P4) is checked by a lookup. A sign pattern on the five points $\{0, \dots, 4\}$ has 20 signs, so there are 2^{20} of them. Before the search, the program lists the 16 patterns that satisfy (i) and (ii) of Lemma 3.5 (two choices of s, and two choices each for the three unrestricted signs) and closes this list under relabelling and reorientation of $\{0, \dots, 4\}$. The result is a set E of 53760 patterns. An injective map $p : \{0, \dots, 4\} \to V$ is a bijection of $\{0, \dots, 4\}$ composed with the increasing identification of $\{0, \dots, 4\}$ with the image of p. So (P4) holds if and only if, for every five-element subset $S \subseteq V$, the restriction of (σ, χ) to S is not in E. This is what the program checks, once for each S, when the last triple inside S is assigned.

5.3. **The result.** For every σ the search finds no χ. This proves Theorem 4.2. The search visits 1 987 999 nodes in total and runs in two seconds on a laptop.

On the use of AI

The ideas for proofs were developed with the help of Anthropic's Claude models. The program was written by the authors and reviewed and tested both by the authors and by the models. The text was proofread and edited with the same models. The authors take full responsibility for the correctness of the results.

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